O-P-1731
October 5, 1973

FEDERAL SPECIFICATION

PENETRATING FLUID

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a penetrating fluid for loosening frozen and corroded metallic parts.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 791/GEN - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 171-190 - Hazardous Materials Regulations of Transportation (Including Specifications for Shipping Containers).

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

FSC 6850

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D92-66 - Flash and Fire points by Cleveland Open Cup.
D97-66 - Pour point.
D445-65 - Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities).
D1218-61 - Refractive Index and Refractive Dispersion of Hydrocarbon Liquids.
E70-68 - pH of Aqueous Solutions with the Glass Electrode.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The penetrating fluid shall be formulated as necessary to meet the requirements of this specification. It shall be non-flammable and contain no mineral oil.

3.1.1 Propellant. The propellant used in the aerosol can shall be composed of one of the following gases: Carbon dioxide, nitrous oxide, nitrogen, or fluorinated hydrocarbons. The propellant shall be capable of propelling all the penetrating fluid in the can and shall be non-flammable as defined in CFR 49, DOT regulations, paragraph 173.300.

3.2 Toxicity. When used for its intended purpose, the penetrating fluid and propellant, when applicable, shall have no adverse effect on the health of personnel. Questions relating to the health effects of the fluid and propellant shall be referred to the appropriate departmental medical service acting as advisor to the procuring agency.

3.3 Physical properties. The penetrating fluid shall conform to test requirements shown in Table I and 3.3.1.

TABLE I. Physical properties

Characteristics	Min	Max
pH at 25°C.	9.3	9.7
Pour Point, Degree, F.	-	27
Refractive index, at 20°C.	1.3516	1.3636
Viscosity, Centistokes at 100°F.	-	1.5
Flash point	None to boiling	

3.3.1 Solubility. The penetrating fluid mixed with equal parts of distilled water shall show no trace of separation, milkiness or turbidity when tested as specified in 4.3.1.

3.4 Aerosol containers (see 6.3). Ten ounces of penetrating fluid shall be packaged in a twelve-ounce aerosol container. The container shall be coated internally to prevent corrosion, and shall be fitted with an activating mechanism and a snap-fitted cover to protect the mechanism. The activating mechanism and spray nozzle shall be designed for convenient operation and shall require use of only one hand to hold and operate the unit. The mechanism shall be readily operated by a downward finger pressure, which shall activate the valve. The spray nozzle shall be so marked

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or designed that the direction of discharge is obvious. The complete unit shall be in compliance with the applicable DOT regulations or other carrier regulations applicable to the mode of transportation.

3.4.1 Valve. The valve shall be centrally located on the valve cup and shall be capable of being operated by the activating mechanism. The valve shall be the wet spray type having a nozzle through which the propellant and active ingredient shall be delivered. The valve shall not leak or clog when tested in accordance with 4.4.3.3.

3.4.2 Pressure. The complete unit shall be capable of withstanding the maximum DOT recommended pressure for the applicable class container, when tested in accordance with 4.4.3.1.

3.4.3 Leakage. The complete unit shall show no evidence of leakage when tested in accordance with 4.4.3.2.

3.5 Workmanship. The penetrating fluid shall be free of dirt, sediment, objectionable odor, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of all penetrating fluid, packaged in the same type containers by one manufacturer and offered for delivery at one time.

4.2.2 Sampling for test. From each lot a random sample of filled containers of penetrating fluid shall be selected in accordance with MIL-STD-105, at inspection level S-2.

4.3 Tests. Penetrating fluid samples selected in accordance with 4.2.2 shall be tested in accordance with the following methods specified in table II and 4.3.1.

TABLE II. Physical tests

Test	ASTM Test Method No.
pH	E-70
Pour point	D-97
Refractive index	D-1218
Viscosity	D-445
Flash point	D-92

4.3.1 Solubility. Use equal portions of penetrating fluid and distilled water to form a composite specimen of not less than one pint. Maintain the solution at 25° + 2°C. and stir vigorously for 5 minutes. Examine visually for compliance with 3.3.1.

4.3.2 Penetrating fluid, in aerosol containers. Upon completion of tests on aerosol containers, penetrating fluid collected in 4.4.3.3 shall be tested in accordance with 4.3 and 4.3.1.

4.4 Aerosol containers inspection.

4.4.1 Inspection lot. For the purpose of sampling, a lot shall consist of all penetrating fluid packaged in aerosol containers and offered for delivery at one time.

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4.4.2 Sampling. A random sample of containers shall be selected from each lot in accordance with MIL-STD-105, at inspection level I. The acceptable quality level shall be 1.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.2 shall be subjected to the following tests. Unless otherwise specified the physical tests shall be made under one atmosphere of pressure and at a temperature of 70° ± 2°F. Fully loaded aerosol containers shall be utilized for each test.

4.4.3.1 Pressure. The aerosol can pressure shall be measured by means of a standard pressure gage fitted with a suitable adapter in accordance with DOT regulations.

4.4.3.2 Leakage. Fully loaded aerosol containers shall be tested for leakage in accordance with DOT regulations.

4.4.3.3 Valve operation. Self-closing valves on aerosol containers shall be manually operated 30 times successively without requiring excessive finger pressure and shall close immediately upon each release. The penetrating fluid shall be propelled in a wet spray or stream and collected in a beaker for tests specified in 4.3. The valve shall then be tested for leakage by submersion for 5 minutes in water containing a wetting agent and maintained at a temperature of 70° ± 2°F.

4.5 Examination for preparation for delivery. An examination shall be made to determine that packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE III. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	More or less than required.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers, cushioning inadequate, improper or missing.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The penetrating fluid in 8-ounce, 12-ounce or 1-gallon quantities, as specified (see 6.2) shall be furnished in containers described in 5.1.1.1, 5.1.1.2, or 5.1.1.3, as appropriate. The penetrating fluid packaged shall neither affect nor be affected by the container.

5.1.1.1 Eight ounces. Eight-ounce quantity of penetrating fluid shall be furnished in a can conforming to PPP-C-96, type V, class 6, plan B or in a polyethylene squeeze bottle (0.95 density minimum) with a leak-proof screw cap, lock top turret closure.

5.1.1.2 Twelve ounces. Twelve-ounce cans specified in 3.4 do not require packaging.

5.1.1.3 One gallon. One gallon of penetrating fluid shall be furnished in a can conforming to PPP-C-96, type V, class 4, plan B. Leak-proof plastic screw caps are acceptable.

5.1.2 Level B. The penetrating fluid shall be packaged as specified in 5.1.1 except that plan A exterior coating is acceptable.

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5.1.3 Level C. The penetrating fluid shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The penetrating fluid, packaged as specified in 5.1 shall be packed in accordance with level A requirements of the appendix to PPP-C-96.

5.2.2 Level B. Twenty-four 8-ounce; twelve 12-ounce or six 1-gallon containers packaged as specified in 5.1 shall be packed in a box conforming to PPP-B-636. Separators, pads or cells of fiberboard conforming to PPP-F-320, type CF, variety SW shall be used. The tiers of containers shall be separated by fiberboard pads.

5.2.3 Level C. The penetrating fluid, packaged as specified in 5.1 in quantities as specified (see 6.2) shall be packed to assure carrier acceptance and safe arrival at destination. Containers shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special markings required by the contract or order and 5.3.1, the interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

5.3.1 Special markings. Special markings for aerosol cans shall be as follows:

WARNING: Contents under pressure. Do not expose to high temperatures. Do not store at temperatures exceeding 120°F. Keep away from direct sunlight, radiators, stoves, hot water, and other sources of heat. Do not puncture or incinerate can.

6. NOTES

6.1 Intended use. Penetrating fluid covered by this specification is intended for use in the freeing of corroded or rusted parts without causing damage. For use also as a rubber lubricant and rubber preservative.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Type of container, required (see 5.1.1 and 5.1.2).
(d) Quantities required (5.2.2).
(e) Marking required (5.3).

6.3 Aerosol cans. Propellant cans may be used where anticipated shipping and storage conditions do not exceed temperatures of 120°F. Propellant cans should not be exposed to direct sunlight, radiators, stoves, hot water, and other sources of heat. It is noted that section 173.306 of DOT regulations specifies that each completed metal container filled for shipment must be heated until content reaches a temperature of 130°F., without evidence of leakage, distortion, or other defects.

MILITARY INTEREST:

Custodians:

Army-MR
Navy-AS
Air Force-68

Review Activities:

Army-MR
Navy-SH, MC
Air Force-68

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GPO
GSA-PBO
NASA-MSF
TVA
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User Activities:

Navy-AS, OS, YD
Air Force-68

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